

1 Effective Stress

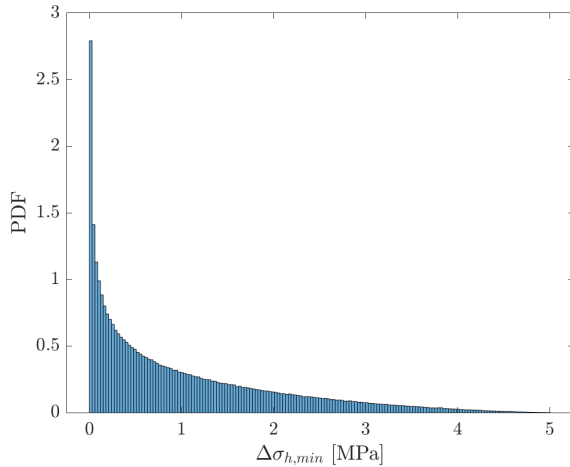
Azizmohammadi & Matthäi (2017): "Since no better data are available, constant far-field stress conditions were applied to the fracture patterns to compute aperture distributions. We assume the common scenario of a shallow burial depth in a normal faulting regime with a $\sigma_1 = 15 \text{ MPa}$ ¹, $\sigma_3 = 10 \text{ MPa}$ and fluid pressure of 9 MPa. ^{a hydrostatic?}"

In the following, we will use $\sigma_{H,\max}$ and $\sigma_{h,\min}$ instead of σ_1 and σ_3 , respectively. Assuming a density of $2.2 \text{ g/cm}^3 = 2200 \text{ kg/m}^3$, the minimum depth which fulfils $\sigma_v > \sigma_{H,\max}$ is 695 m.

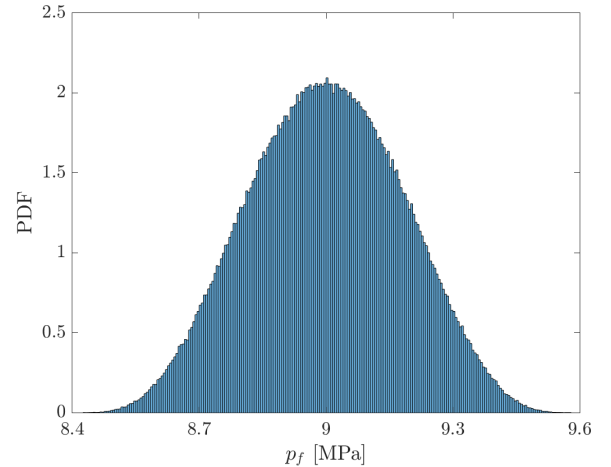
The uncertainty in the far-field stress is relatively small. Therefore, we use the same far-field stress for all realisations. However, the local stress can deviate from the far-field stress. Concretely, the differential stress can be smaller² and the principal stresses can be rotated. Therefore, we assign some uncertainty to the orientation of the maximal principal horizontal stress β and to the magnitude of the least principal stress $\sigma_{h,\min}$ while keeping $\sigma_{H,\max}$ constant.

The fluid pressure is larger or equal to the hydrostatic pressure. A hydrostatic pressure of 9 MPa corresponds therefore to a depth of 917 m or less. Since the Barton & Bandis model cannot deal with tensile opening, the fluid pressure must be smaller than the fracture normal stress (i.e. $\sigma_{eff} > 0$). Therefore, the fluid pressure must be smaller than $\sigma_{h,\min}$. In our framework, every realisation has a slightly different fluid pressure while the fluid pressure is the same for all fractures within one realisation.

Equation	Uncertainty ($\mu \pm \sigma$)
1) $\theta = \arccos(n_1 \cos \beta + n_2 \sin \beta)$	$\beta = 0^\circ \pm 5^\circ$
2) $\sigma_n = \sigma_{H,\max} \cos^2 \theta + \sigma_{h,\min} \sin^2 \theta$	$\sigma_{h,\min} = 10 \text{ MPa} + \Delta\sigma_{h,\min}$ $\Delta\sigma_{h,\min}$ from scaled β -distribution (see Fig. 1a) such that $\Delta\sigma_{h,\min} \in [0, 5] \text{ MPa}$, $\mu = 1 \text{ MPa}$ and $\sigma = 1 \text{ MPa}$ $\sigma_{H,\max} = 15 \text{ MPa}$
3) $\sigma_s = (\sigma_{h,\min} - \sigma_{H,\max}) \sin \theta \cos \theta$	
4) $\sigma_{eff} = \sigma_n - p_f$	$p_f = 9 \text{ MPa} \pm 2\%$ (constant in one realisation)



(a) Least principal stress



(b) Fluid pressure

Figure 1

¹Actually, σ_1 should here be called σ_2

²In a 2D domain, the differential stress is $\sigma_{H,\max} - \sigma_{h,\min}$

2 Normal Closure

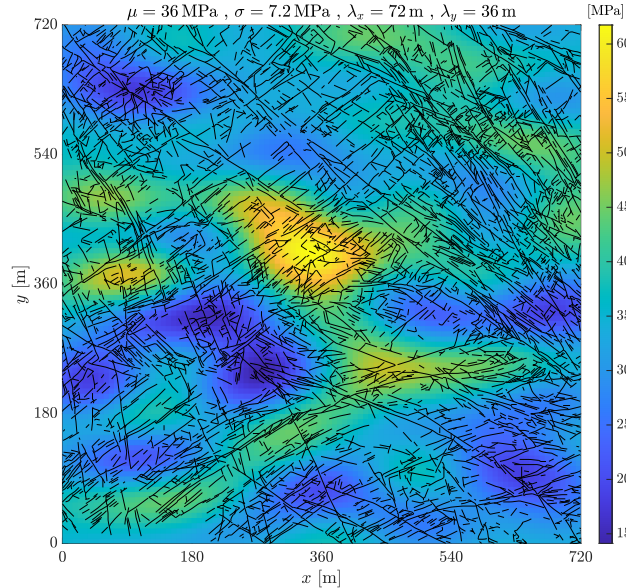
Odling (1997): "The fracture system studied occurs in Devonian-age sandstones of Hornelen in western Norway [...]. In Hornelen, sandstones and conglomerates fill a fault-bound basin [...].". In a simplified tectonic map, Hartz et al. (2002) shows the Hornelen area as Devonian collapse basin (alluvial and fluvial deposits).

The joint roughness coefficient JRC is one of the main sources of uncertainty. The joint roughness coefficient varies from 0 for perfectly smooth fractures to approximately 20 for rough undulating fractures (Barton, 1973). We assume that in our fractured reservoir $JRC \in [5, 20]$ and that longer fractures tend to have larger JRC . Barton et al. (1985) suggest formulas to take the size effect of JRC (and also JCS) into account. However, those formulas shouldn't be applied if the fracture length is several orders of magnitude ^{above} higher than the lab scale $L_0 \approx 0.1$ m, as it is in our case. Furthermore, JRC decreases with fracture length which is not what we expect. I think those formulas only model perfectly straight fractures, but we want to include large-scale roughness (i.e. curved or undulated fractures) in the JRC values as well.³ We assume that JRC is constant over the length of a fracture, i.e. the value is the same for all segments of a fracture.

We set joint wall compressive strength equal to unconfined compression strength (i.e. $JCS = \sigma_c$). This corresponds to the case of an unaltered and unweathered joint. The unconfined compression strength is a material property of the rock. Material properties vary gradually throughout the field. We model the parameter as a Gaussian random field from which we sample its value at the fracture locations. Assuming that the rock type is a porous quartz-sandstone, the unconfined compression strength is between 33 MPa to 38 MPa for saturated rock and up to 83 MPa for oven-dried samples (Ojo & Brook, 1990). The values given in (Barton, 1973) are slightly smaller. Hawkins & McConnell (1991) studies a wide range of different sandstones where $\sigma_c \in [10 \text{ MPa}, 300 \text{ MPa}]$ ⁴.

Bandis et al. (1983) propose an empirical relation for the maximum closure and they provide uncertainty for the four model parameters. However, when sampling these parameters from uncorrelated normal distributions with corresponding mean and standard deviation, a maximum closure larger than the initial aperture or even negative values for the maximum closure frequently occur. We therefore use an acceptance-rejection sampling method where we reject any $v_m < 0$ or $v_m > a_0$.

Unfortunately, no uncertainty was provided in Bandis (1980) for the model parameters of the empirical equation for the initial normal stiffness. Therefore, we add some uncertainty to the resulting value.



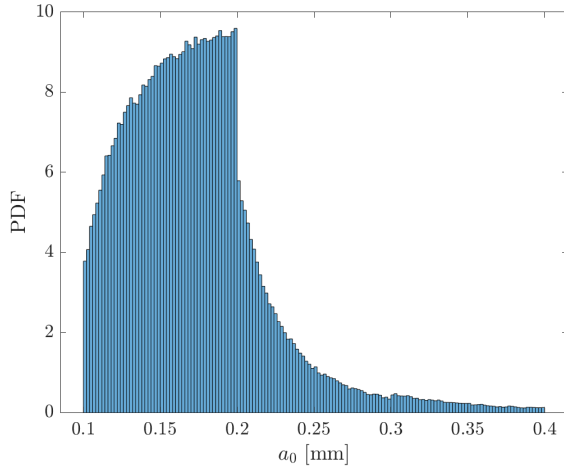
(a) Unconfined compression strength

Figure 2

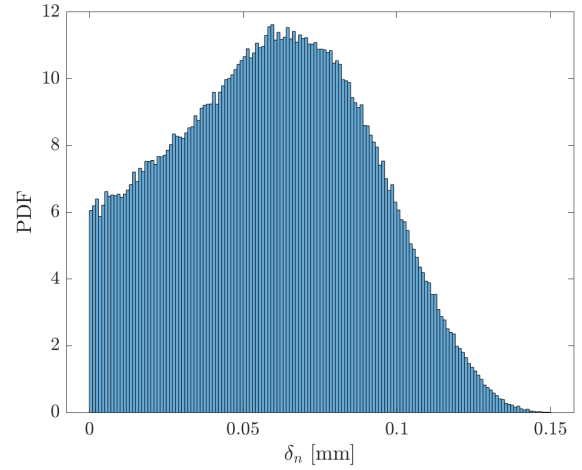
³Does this make sense?

⁴Which ones are quartz sandstones?

Equation	Uncertainty ($\mu \pm \sigma$)
5) $a_0 = \frac{JRC}{5} \left(0.2 \frac{\sigma_c}{JCS} - 0.1 \right) = \frac{JRC}{50}$	JRC as a function of fracture length Shortest fractures ($L \leq 10$ m): $JRC \sim \mathcal{U}[5, 10]$ Longest fractures ($L \geq 100$ m): $JRC \sim \mathcal{U}[15, 20]$ Linear interpolation of bounds in between $\sigma_c = 36 \text{ MPa} \pm 20\%$ (see Fig. 2a) Gaussian RF with Gaussian correlation function Correlation length $\lambda_x = 72 \text{ m}$, $\lambda_y = 36 \text{ m}$ $JCS = \sigma_c$
6) $v_m = A + B \cdot JRC + C \left(\frac{JCS}{a_0} \right)^D$	$A = -0.1032 \pm 0.0680$ (Bandis et al., 1983) $B = -0.0074 \pm 0.0039$ $C = 1.1350 \pm 0.3261$ $D = -0.2510 \pm 0.1029$ Use acceptance-rejection sampling to ensure $v_m \in [0, a_0]$
7) $K_{ni} = -7.154 + 1.748 \cdot JRC + 0.0178 \left(\frac{JCS}{a_0} \right)$	$K_{ni} = K_{ni} \pm 5\%$
8) $\delta_n = \frac{\sigma_{eff} v_m}{K_{ni} v_m + \sigma_{eff}}$	



(a) Initial aperture



(b) Normal closure

Figure 3

3 Shear Displacement

The model of Barton & Bandis is not suitable ^{for calculating} to calculate shear displacement. When it is implemented in a FEM code, shear displacement along with local normal and shear stresses are calculated by the mechanic solver. In this work, we have to estimate the shear displacement from the far-field stresses and the properties of fracture and material. We use a model introduced by Willis-Richards et al. (1996) and Rahman et al. (2002) which uses the excess shear stress $\Delta\tau$ and fracture shear stiffness K_s .

The residual friction angle is another important source of uncertainty. So far, I haven't found any values in the literature. In one of our meetings, we defined $\phi_r \in [20^\circ, 35^\circ]$ so I used a normal distribution with mean 27.5° and standard deviation 2.5° .

The material properties E and ν are modelled similarly to the unconfined compression strength with a Gaussian random field. Lei & Gao (2019) assume a Poisson's ratio of $\nu = 0.25$ and a mean Young's modulus of $E = 20$ GPa in their simulations of the $18\text{ m} \times 18\text{ m}$ outcrop map of the Hornelen Basin. Hart & Wang (1995) measured the value for Berea sandstone (a graywacke) and obtained $E \in [13\text{ GPa}, 28\text{ GPa}]$ and $\nu \in [0.12, 0.23]$ for drained and $E \in [10\text{ GPa}, 19\text{ GPa}]$ and $\nu \in [0.26, 0.40]$ for undrained conditions. In Hawkins & McConnell (1991), the Young's modulus is in the range $E \in [1\text{ GPa}, 40\text{ GPa}]$ and the Poisson's ratio $\nu \in [0.09, 0.4]$ for most sandstones.⁵ We use the values from Lei & Gao (2019) and add some uncertainty.

We use the general solution for shear stiffness of a fracture proposed by Dietrich (1992). For a penny-shaped circular crack $C_g = 7\pi/24 = 0.92$ (Eshelby, 1957). Because we don't have penny-shaped fractures, we set $C_g = 1$ and increase the uncertainty in other parameters. We define r as half of the total length of the fracture.

The Young's modulus and to a lesser extent also the Poisson's ratio have a direct impact on the fracture shear stiffness K_s through the shear modulus G . They have therefore a great influence on the shear displacement and consequently on the shear dilation. The values from Lei & Gao (2019) give rather high shear stiffness and consequently quite small shear displacements (Fig. 5b).⁶

Here, we assume that shear displacement is constant over the fracture length. We neglect that shear stiffness is minimal and shear displacement maximal in the middle of a fracture where the latter tends towards zero at the fracture tips. More information about this, including the Cruikshank model, can be found e.g. in Milliotte et al. (2018).

Equation	Uncertainty ($\mu \pm \sigma$)
9) $\Delta\tau = \sigma_s - \sigma_{eff} \cdot \tan(\phi_r)$	$27.5^\circ \pm 2.5^\circ$ which makes sure almost all values are in $\phi_r \in [20^\circ, 35^\circ]$
10) $G = \frac{E}{2(1+\nu)}$	$E = 20\text{ GPa} \pm 15\%$ (see Fig. 4a) Gaussian RF with Gaussian correlation function Correlation length $\lambda_x = 72\text{ m}$, $\lambda_y = 36\text{ m}$ $\nu = 0.25 \pm 3\%$ (see Fig. 4b) Gaussian RF with Gaussian correlation function Correlation length $\lambda_x = 72\text{ m}$, $\lambda_y = 36\text{ m}$
11) $K_s = C_g \frac{G}{r}$	$C_g = 1$ $r = \frac{L}{2}$ where $L = \sum L_i$
12) $\delta_s = \frac{\Delta\tau}{K_s}$ if $\Delta\tau > 0$, else $\delta_s = 0$	

4 Shear Dilation

We calculate peak shear displacement according to Asadollahi & Tonon (2010) (Fig. 5a). An alternative would be to use the formula by Barton (1982). We then use Fig. 6a to calculate the mobilized joint roughness coefficient (Fig. 6a). Fig. 6b shows the integrated shear dilation over shear displacement. Since $JRC_{mob} < 0$ for $\delta_s < 0.3\delta_{peak}$, the shear dilation is negative for shear displacement up to around peak shear displacement. The negative shear dilation can be greater than the initial aperture, which leads to negative fracture apertures.⁷

⁵ Again which sandstone is closest to the one from Hornelen?

⁶ What can we do about it?

⁷ How should we deal with this? Shall we define a minimum fracture aperture of e.g. $1\text{ }\mu\text{m}$?

This is a strange one and maybe related to an error in the paper. Are there any errata published?

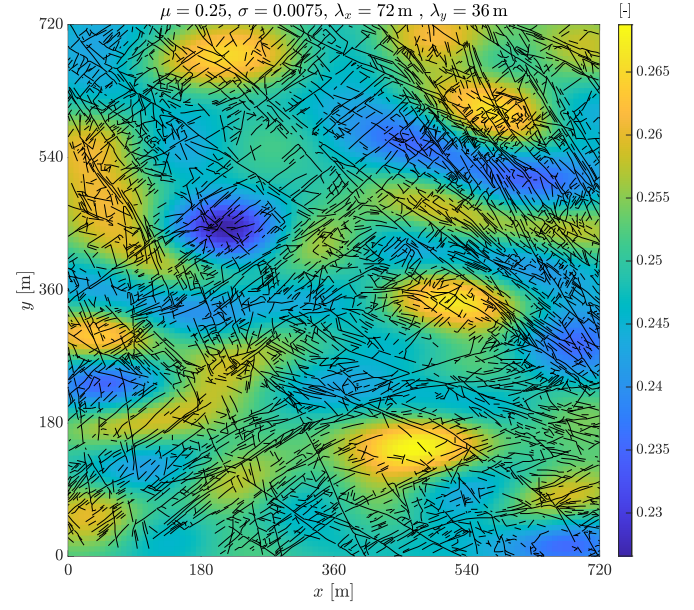
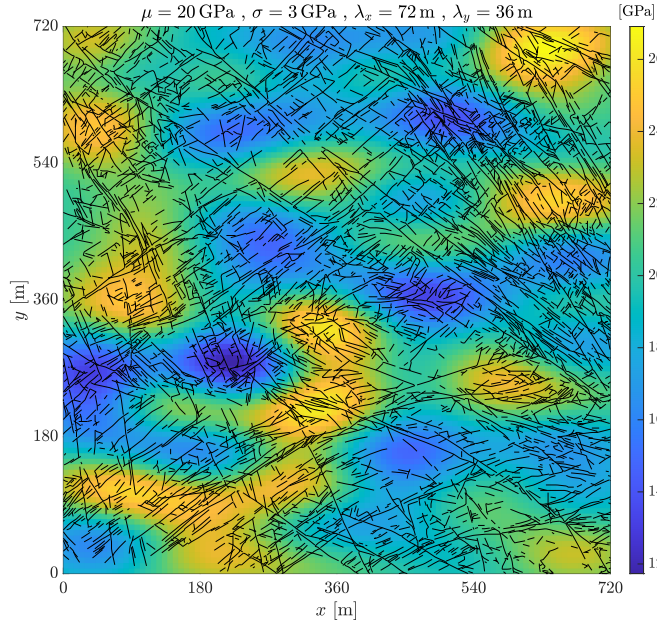


Figure 4

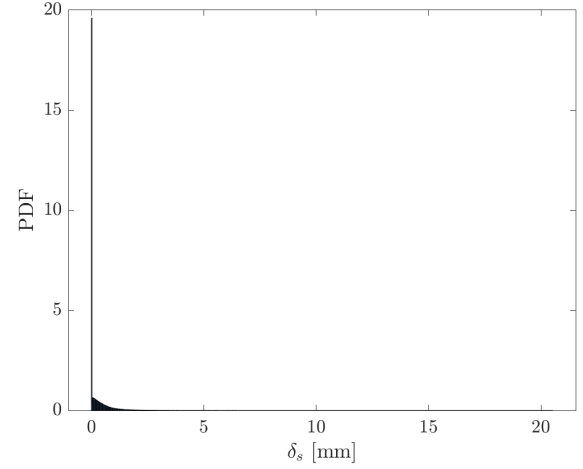
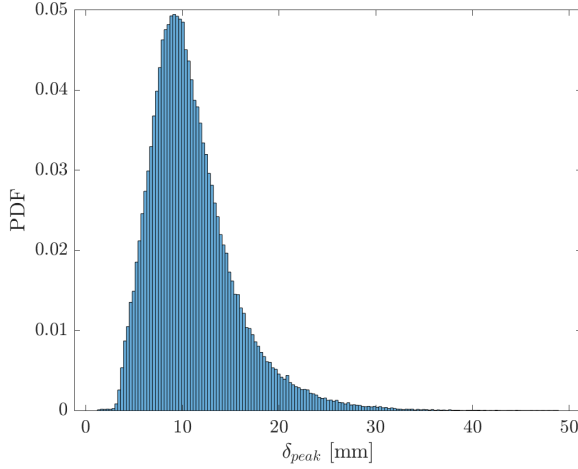


Figure 5

Barton & Choubey (1977) propose to use a damage coefficient M which accounts for damage of the asperity at high normal stresses by decreasing the mobilized dilation angle. $M = 1$ corresponds to shearing under low normal stress where no damage occurs while $M = 2$ models asperity damage for shearing under high normal stress. When using Equation (15) as proposed by Barton & Choubey (1977), we obtain $M > 2$ for some fractures due to their high JRC values (Fig. 7a).⁸

At this stage, we don't include any additional uncertainty.

⁸Is this a problem?

Equation

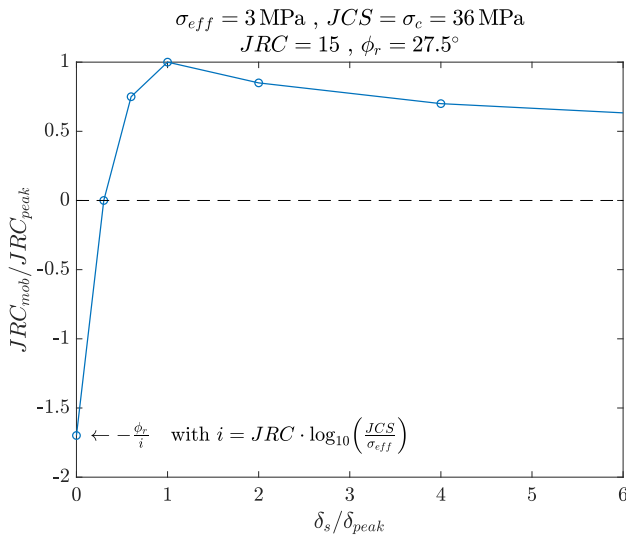
$$13) \quad \delta_{peak} = 0.0077L^{0.45} \left(\frac{\sigma_{eff}}{JCS} \right)^{0.34} \cos \left(JRC \cdot \log_{10} \left(\frac{JCS}{\sigma_{eff}} \right) \right)$$

$$14) \quad JRC_{mob} = f(JRC, \delta_{peak}, \delta_s) \rightarrow \text{diagram Fig. 6a}$$

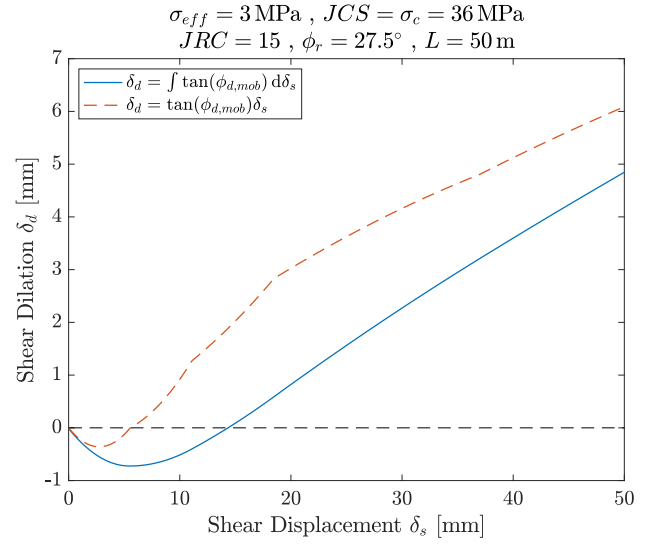
$$15) \quad M = 0.7 + JRC_n \left/ \left[12 \log_{10} \left(\frac{JCS_n}{\sigma_{eff}} \right) \right] \right.$$

$$16) \quad \phi_{d,mob} = \frac{1}{M} JRC_{mob} \log_{10} \left(\frac{JCS_n}{\sigma_{eff}} \right)$$

$$17) \quad d\delta_d = d\delta_s \tan(\phi_{d,mob}) \rightarrow \delta_d = \int_0^{\delta_s} \tan(\phi_{d,mob}) d\delta_s$$

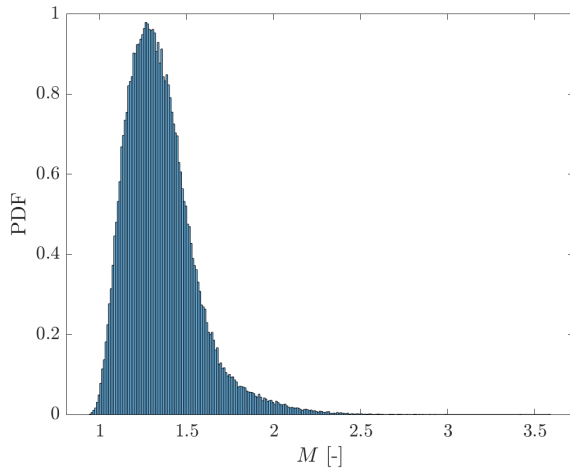


(a) Mobilized joint roughness coefficient after Barton et al. (1985)

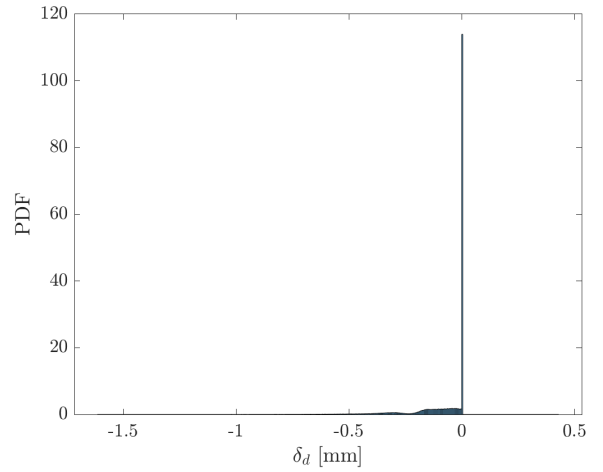


(b) Shear dilation

Figure 6



(a) Damage coefficient



(b) Shear dilation

Figure 7

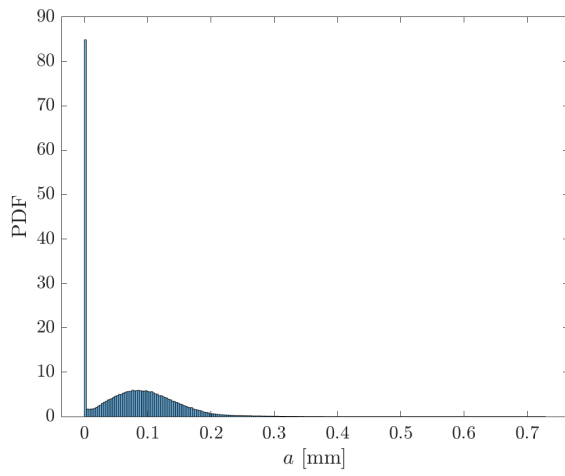
5 Aperture

In the model of Barton & Bandis, the ~~here~~ calculated aperture a is actually the mechanical aperture. They provide a formula to calculate the hydraulic aperture from the mechanical aperture. We neglect this and set the hydraulic aperture equal to the mechanical aperture.⁹

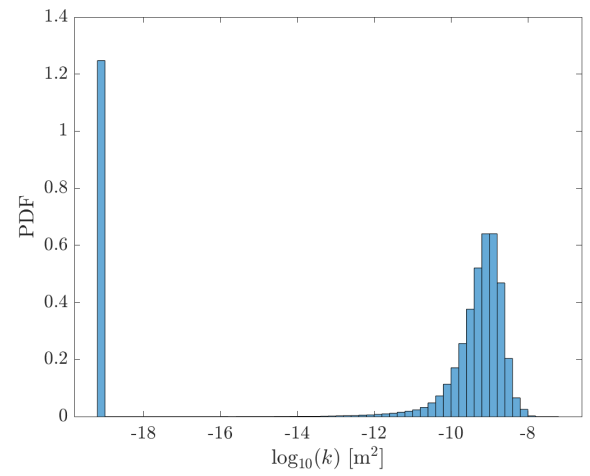
Equation

18) $a = a_0 - \delta_n + \delta_d$

19) $k = a^2/12$



(a) Fracture aperture



(b) Fracture permeability

Figure 8

⁹Is this a reasonable simplification?

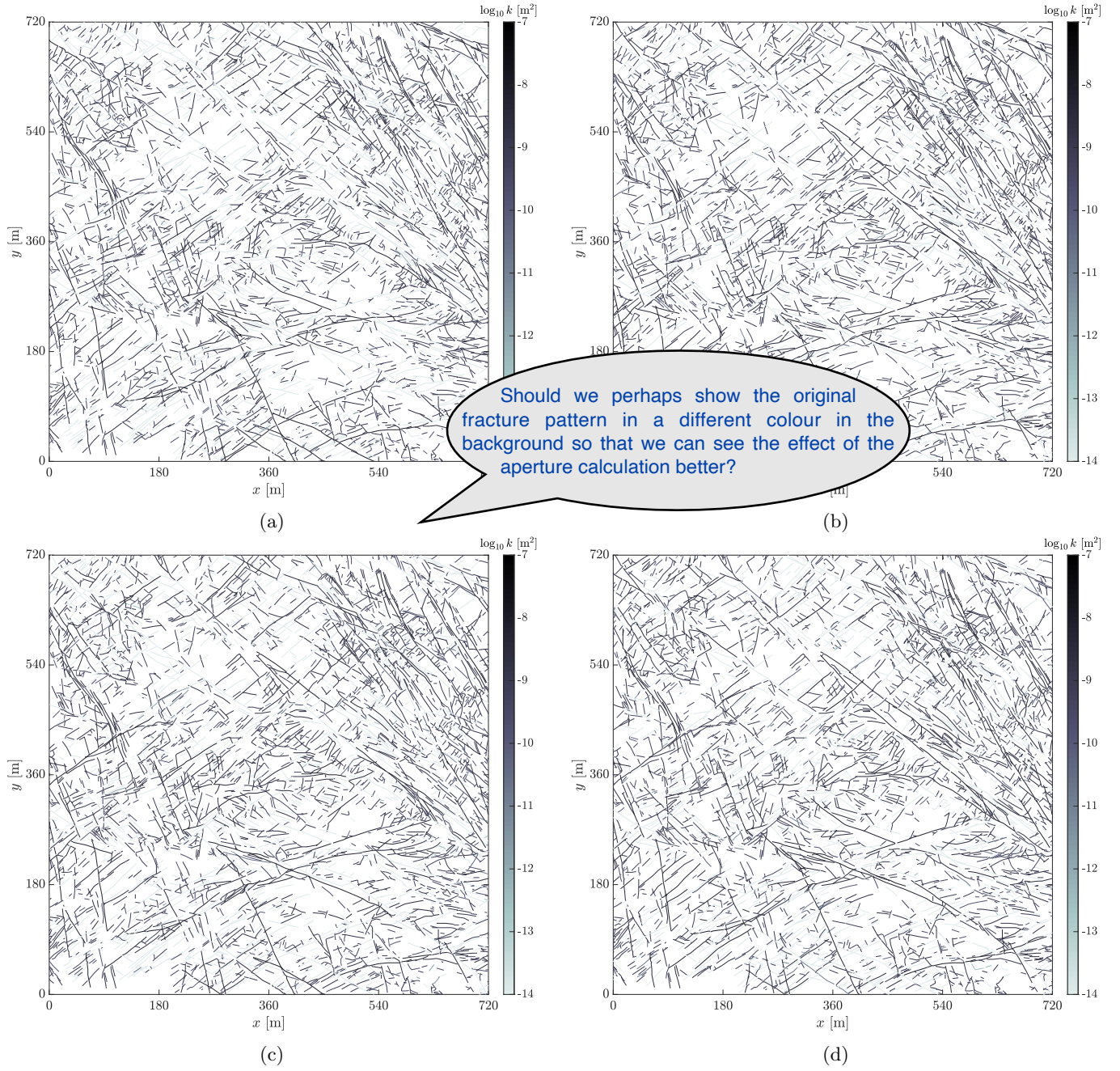


Figure 9: Four different realisation of the fracture aperture. Line thickness is proportional to aperture while line color is proportional to $\log_{10}$ of the permeability.

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